



Indywidualny Plan Badawczy (IPB) Individual Research Plan (IRP)

1. Podstawowe dane / Basic data

Podstawowe informacje dotyczące doktoranta i realizowanej rozprawy doktorskiej. W przypadku promotora spoza Politechniki Wrocławskiej należy podać afiliację. / *Basic information on the doctoral student and the doctoral dissertation underway. In the case of a supervisor from outside Wrocław University of Science and Technology, affiliation should be provided.*

Imię i nazwisko / Name, surname	Grzegorz Piotrowski
Dyscyplina kształcenia / Discipline of education	informatyka techniczna i telekomunikacja / information and communication technology
Wydział / Faculty	W4 Wydział Informatyki i Telekomunikacji / Faculty of Information and Communication Technology
Katedra / Department	K46 Katedra Sztucznej Inteligencji / Department of Artificial Intelligence
Data rozpoczęcia kształcenia / Date of commencement of the doctoral education	01.10.2025
Nr ORCID	0009-0004-6013-0461
Promotor/ promotorzy / Supervisor(s)	dr hab. inż. Tomasz Jan Kajdanowicz, prof. uczelni, Politechnika Wrocławska (K46, W4), ORCID 0000-0002-8417-1012
Promotor pomocniczy / Assistant supervisor	—

2. Temat rozprawy doktorskiej / Topic of doctoral dissertation

Propozycja tematu rozprawy doktorskiej. Przedstawiony temat nie musi stanowić ostatecznego tytułu, jaki zostanie nadany rozprawie doktorskiej złożonej na zakończenie kształcenia w Szkole Doktorskiej. / *Proposed topic for a doctoral dissertation. The presented topic does not have to constitute the final title of the doctoral dissertation submitted at the end of education at the Doctoral School.*

Metody uczenia reprezentacji dla generalizacji modeli uczenia głębokiego między symulacją a rzeczywistością w fizycznej sztucznej inteligencji / Representation learning methods for simulation-to-reality generalization of deep learning models in physical AI

3. Harmonogram przygotowania rozprawy doktorskiej / Schedule for the preparation of the doctoral dissertation

Harmonogram przygotowania rozprawy doktorskiej z wyszczególnieniem etapów i miejsc prowadzonych badań, w tym terminów zakończenia badań cząstkowych (semestr kształcenia) i opracowania ich wyników oraz opcjonalnie w zależności od rodzaju prowadzonych badań, wybranych między innymi spośród następujących:

- realizacja badań (w tym przygotowanie stanowiska badawczego, przygotowanie materiałów do badań, wykonanie pomiarów/eksperymentów, opracowanie otrzymanych wyników),
- przygotowanie i wysłanie artykułu naukowego do redakcji czasopisma naukowego,



- wysłanie zgłoszenia patentowego do właściwego urzędu,
- opracowanie propozycji wdrożenia dla przemysłu,
- udział w konferencjach, warsztatach lub szkołach letnich/zimowych i prezentacja wyników badań,
- przygotowanie wniosku i wystąpienie o grant,
- mobilność krajowa lub zagraniczna doktoranta (w tym konsultacje naukowe i wizyty studyjne w ośrodkach krajowych lub zagranicznych).

A schedule for the preparation of the doctoral dissertation detailing the stages and locations of the research to be conducted, including the dates for completing the sub-studies and developing their results, and optionally, depending on the type of research to be conducted, selected among the following:

- *execution of the research (including preparation of the research site, preparation of materials for the research, execution of measurements/experiments, elaboration of the obtained results),*
- *preparation and submission of a scientific article to the editor of a scientific journal,*
- *submission of a patent application to the relevant office,*
- *development of an implementation proposal for the industry,*
- *participation in conferences, workshops or summer/winter schools and presentation of research results,*
- *preparation of an application and application for a scholarship,*
- *domestic or foreign mobility of the doctoral student (including scientific consultations and study visits to domestic or foreign centres).*

Okres realizacji zadania (semestr kształcenia) / Task implementation period (semester)	Zwięzły opis zadania / Brief description of the task
Semestr 1	1. Participation in the Doctoral School courses. 2. Getting acquainted with the research carried out in the Department of Artificial Intelligence and choosing the research area together with the supervisor.
Semestr 2	1. Participation in the Doctoral School courses. 2. Specifying the topic of the doctoral dissertation and preparing the Individual Research Plan.
Semestr 3	1. Literature review covering simulation-to-reality transfer, domain randomization and domain adaptation, neural scene reconstruction from real data (such as 3D Gaussian Splatting) and deep learning for robot navigation and manipulation, including participation in the internal reading group. 2. Identification of the most important research gaps in the generalization of deep learning models trained in simulations built from real data. 3. Preparation of the research environment for robot navigation and robotic manipulation, with shared reconstruction tools and computing resources. For navigation, it includes real scene captures, digital twins in a navigation simulator and a dataset-based proxy that measures the reconstruction-fidelity gap. For manipulation, it includes digital twins of real tabletop scenes built from images of public robot datasets and paired with published real-robot results of the same policies. 4. Starting the research on which reconstruction errors of a digital twin, which vary across a scene, determine the generalization of navigation and manipulation models trained in it.



Semestr 4	1. Continuation of the literature review, including participation in the internal reading group that discusses new methods of simulation-to-reality transfer and physical AI. 2. Finishing the research on harmful reconstruction errors in both tasks and on randomizing each region of a digital twin according to its estimated reconstruction error and task relevance. 3. Writing a scientific article for a conference from the ministerial list on the reconstruction errors that determine generalization in navigation and manipulation, planned for submission to a robotics conference or journal from the ministerial list focused on physical AI, e.g. ICRA 2028 or IEEE Robotics and Automation Letters (RA-L). 4. Starting the research on localizing the simulation-to-reality gap inside navigation and manipulation models, with first results on the stage at which task information is lost on real inputs. 5. Preparation of the Preludium research grant proposal to the National Science Centre. 6. Participation in an international scientific conference.
Semestr 5	1. Continuation of the literature review, including participation in the internal reading group that discusses new methods of simulation-to-reality transfer and physical AI. 2. Finishing the research on localizing the simulation-to-reality gap inside the models of both tasks and on aligning representations at that stage. 3. Writing a scientific article for a conference from the ministerial list on localizing the simulation-to-reality gap inside deep models, e.g. ICML 2028 or CVPR 2028. 4. Research on selecting real data under a limited budget to correct both the digital twin and the model in both tasks. 5. Starting the research on predicting whether the improvements transfer to unseen environments and between navigation and manipulation.
Semestr 6	1. Continuation of the literature review, including participation in the internal reading group that discusses new methods of simulation-to-reality transfer and physical AI. 2. Finishing the research on selecting real data and on predicting the transfer of improvements across environments and tasks, complemented by controlled studies of physical parameters in a simulator with hidden parameters. 3. Writing a scientific article for a conference from the ministerial list on selecting real data under a limited budget and predicting transfer, e.g. NeurIPS 2028. 4. Initial validation of the developed methods on a real robot, if access to a robotics laboratory of Wrocław University of Science and Technology allows. 5. Research internship in a foreign research group working on robot learning, if funding allows.



Semestr 7	1. Continuation of the literature review, including participation in the internal reading group that discusses new methods of simulation-to-reality transfer and physical AI. 2. Finishing the experiments and, if possible, the validation on a real robot. 3. Writing a scientific article summarizing the results for a conference or journal from the ministerial list. 4. Participation in an international scientific conference. 5. Preparing the structure and outline of the doctoral dissertation.
Semestr 8	Editing the doctoral dissertation and preparing the final version of the document.

4. Termin złożenia rozprawy doktorskiej / Date of the doctoral dissertation submission

Planowany termin złożenia rozprawy doktorskiej nie może przekraczać daty zakończenia ósmego semestru kształcenia w Szkole Doktorskiej. / A planned date for submission of the dissertation, which must not exceed the date of completion of the eighth semester of education in the Doctoral School.

September 2029

(rekomendowana data to ostatni dzień z okresu 4 lata kształcenia w Szkole Doktorskiej, np. jeżeli kształcenie rozpoczęło się 1.10.2025, to proszę wpisać 30.09.2029 /

The recommended date, which marks the final day of the four-year doctoral training period. For example, if your training began on October 1st, 2025, please enter September 30th, 2029)

5. Uzasadnienie wyboru tematu rozprawy doktorskiej / Justification for choosing the topic of the doctoral dissertation

Opis zawierający uzasadnienie wyboru tematu rozprawy doktorskiej uwzględniający potencjalne obszary zastosowania planowanych wyników rozprawy doktorskiej (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description containing the justification for the selection of the topic of the doctoral dissertation, taking into account the potential areas of application of the planned results of the doctoral dissertation (at most 1 page, font 11, line spacing 1).

Deep learning has made great progress in computer vision and natural language processing, largely thanks to vast amounts of data collected from the web. Physical AI, i.e., AI in robots and other embodied systems that perceive and act in the real world, cannot rely on such data. Its models need interaction data, which are slow, expensive and sometimes unsafe to collect with real robots, so they are commonly trained in simulation. Recently, digital twins, i.e., simulations built from real data, have become a promising alternative to hand-crafted simulators. Neural scene reconstruction methods, such as 3D Gaussian Splatting, turn a short recording of a real place into a photorealistic digital twin in which a model can be trained at scale. Real data are turned into a simulation, a model is trained there and then transferred back to reality, which forms a real-to-simulation-to-real loop.

However, the value of this loop depends on how well the trained models generalize. A model that performs well in simulation often fails in reality, since no simulation reproduces the real world exactly: appearance, geometry, lighting and physics are only approximated, and reconstruction errors differ between regions of a scene. The resulting drop in performance is called the simulation-to-reality gap. It is caused by the shift between the distributions of simulated and real data, so it is a particular case of learning under distribution shift and one of the main obstacles to physical AI. The methods proposed so far have several limitations. Domain randomization, even when fitted to real



data, varies a few global parameters regardless of where the simulation is inaccurate, and invariant representations may discard information needed for the task. Moreover, the few real data used for adaptation usually correct either the model or the simulation, but not both. Finally, most methods are evaluated on a single task and a few scenes, and it is not known which properties of the digital twin and of the learned representations determine whether a model generalizes to reality.

Therefore, the dissertation aims to improve the generalization of deep learning models in real-to-simulation-to-real transfer with methods that analyse and align the models' learned representations, applied at the points of the loop where the simulation-to-reality gap arises. The research shall focus on understanding which reconstruction errors of a digital twin are harmful to generalization, on localizing the stage of a model at which task information is lost on real inputs and reducing the gap there, and on choosing, under a fixed budget, the small amount of real data that best corrects both the digital twin and the model. It shall also verify whether the obtained improvements carry over to unseen environments and from one physical task to another, and whether properties of the shift measured beforehand predict when they do. The methods shall not be tied to a particular task; robot navigation and robotic manipulation are planned as two equal testbeds. For navigation, a second, independent capture in public recordings of real scenes shall measure the reconstruction-fidelity gap, and, if access allows, a real robot shall check the direction of the results and the validity of this proxy. For manipulation, digital twins of real tabletop scenes shall be built from images of public robot datasets and compared with published real-robot results of the same policies. Apart from more reliable models, the research could help explain why models trained on synthetic data succeed or fail on real data, which is relevant to machine learning in general.

The results could be applied in service, logistics and assistive robotics, where a robot could be trained in a digital copy of a new warehouse, hospital or home and then operate there reliably after little real-world adaptation. They also concern other domains in which models are trained on simulated or reconstructed data and deployed on real sensor data, such as autonomous driving or inspection robots.

6. Zarys aktualnego stanu badań w tematyce rozprawy doktorskiej / **Outline of the current state of research in the topic of the doctoral dissertation**

Przegląd literaturowy zawierający opis aktualnego stanu badań w tematyce rozprawy doktorskiej uwzględniający odwołania do najważniejszych pozycji literaturowych związanych z tematyką rozprawy doktorskiej (maks. 2 strony, czcionka 11 pt., interlinia 1). / A literature review containing a description of the current state of research in the subject of the doctoral dissertation, including references to the most important literature items related to the subject of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

Deep learning models for navigation and manipulation need more experience than real robots can provide, so they are usually trained in simulation. However, performance in simulation can poorly predict real-world performance unless the simulator is carefully tuned [1]. The simulation-to-reality gap is a case of learning under distribution shift. Domain adaptation theory bounds the error of a model on the target domain by its error on the source domain, the discrepancy between the two distributions, and the joint error of the best single model on both domains [2]. Taken only as motivation, the bound suggests making the training distribution cover reality, making the model insensitive to the remaining discrepancy and, since invariance does not target the joint error and can even increase it, correcting the simulation towards reality, which can reduce this error directly.

Domain randomization varies simulator parameters such as textures so that reality appears as one more variation [3]. Overly wide randomization leads to conservative behaviour, so later works shape its distribution by fitting it to a few real rollouts [4] or by maximizing its entropy while preserving task success [5]. These methods tune a few global parameters of a synthetic simulator and do not address errors that vary across a scene.



Invariance is another option. Domain-adversarial training aligns features across domains [6]. In robotics, aligning the joint distributions of observations and actions in simulated and real data improved co-trained policies, i.e., models that map observations to actions, in reality [7]. However, invariance with low source error does not guarantee transfer, as it can increase the joint error term when the label distributions, here the actions, differ between domains [8], and full invariance may discard task information. Effective simulation-and-real co-training aligns the two domains while keeping them distinguishable [9], so domain separability alone does not show where the gap harms the task. Invariance is also usually enforced at locations chosen in advance, such as the input, the final features or fixed layers, rather than where the gap is measured to arise.

Digital twins are more recent. 3D Gaussian Splatting [10] reconstructs photorealistic scenes from multi-view photographs. In manipulation, twins have been used to train policies from a scan of the target scene [11], to transfer image-based policies to reality without real-world demonstrations [12] and to guide real-world reinforcement learning from a phone capture [13]. Paired simulated and real evaluations showed that scenes matched to real images and controllers of two robot setups track the real-world performance of the same policies [14]. In navigation, policies have been fine-tuned in twins built from a phone capture of a room [15] and trained in twins from monocular urban videos [16]. The effect of simulation factors has been studied mainly globally, e.g., randomization, photorealism and physics realism in real manipulation [17]. For whole workspaces, pretrained feature similarity between rendered and real views predicted the agreement of simulated and real performance better than LPIPS or PSNR [18]. For both tasks it remains open which reconstruction errors that vary across a scene harm generalization, and how training variation should follow the reconstruction error and task relevance of each region.

It is also unclear where inside a model the gap arises. Probing, i.e., reading out information from hidden states with simple classifiers, showed that action fine-tuning degrades the visual representations of vision-language-action models [19]. Separately, fine-tuning selected layers showed that the best layers to adapt depend on the type of shift [20]. To our knowledge, in models trained in twins, the stage at which task information available in simulation stops being recoverable from real inputs has not been located.

The remaining gap is reduced with little real data. Co-training with a small real set yields large gains [21], and active system identification designs, in simulation, an exploration policy whose real trajectory is most informative about physical parameters [22]. The closest work uses a twin to find failure-prone configurations for real rollouts [13]. They use the selected real data to correct either the model or the simulation. To our knowledge, correcting both from the same selected data has not been studied.

The gap can also be decomposed into factors of variation whose difficulty ordering is largely consistent between simulation and reality [23]. It is rarely tested whether measurements taken before a method is applied predict, better than simple indicators such as the size of the gap, whether its benefit carries over to a new scene or to another task.

References

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- [2] Ben-David, S., et al. (2010). A theory of learning from different domains. Machine Learning.
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- [22] Memmel, M., et al. (2024). ASID: Active exploration for system identification in robotic manipulation. ICLR.
- [23] Xie, A., et al. (2024). Decomposing the generalization gap in imitation learning for visual robotic manipulation. ICRA.

7. Pytania i hipotezy badawcze / Research questions and hypotheses

Opis najważniejszych pytań i hipotez badawczych związanych z realizowaną rozprawą doktorską (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description of the most important research questions and hypotheses related to the doctoral dissertation (at most 1 page, font 11, line spacing 1).

The research in the proposed doctoral dissertation addresses the following research questions, each on two physical AI tasks: robot navigation and robotic manipulation.

1. **Which reconstruction errors of a digital twin, varying across a scene, determine how well navigation and manipulation models trained in it generalize?** Twin errors in appearance, geometry and lighting differ across a scene. *Hypothesis:* The performance drop caused by each type of reconstruction error in a region is predicted better by the feature distance of a fixed pretrained visual encoder between held-out views rendered with and without the error than by the error size in image or geometry, e.g., PSNR, LPIPS or depth error. Separately, randomizing the appearance and geometry of each region with a strength growing with its estimated reconstruction error and task relevance generalizes better than uniform randomization of equal total strength. To test this, one error type at a time will be corrected in selected regions, e.g., geometry from a laser scan, or injected into a high-fidelity reference. If harm follows error size, this will be reported and the third question will rely on reconstruction uncertainty and action sensitivity, although uncertainty misses consistently reproduced errors.
2. **At which stage of a model trained or fine-tuned in a twin do real inputs first lose task information available for simulated inputs, and does aligning representations at that stage reduce the simulation-to-reality gap?** *Hypothesis:* In both tasks, there is an identifiable earliest stage, or a narrow range of stages, at which a linear probe trained on simulated hidden states loses accuracy on paired real ones well beyond a control-task baseline, and the lost information



influences the actions. Fine-tuning the layers up to that stage with a loss matching real to paired simulated representations from a few views reduces the gap more than the same loss at the input, the final features, all stages or layers chosen by existing criteria, under equal data and effort. Probes with control tasks will locate this stage, and patching the probed subspace from simulated into real inputs will test whether the model uses this information.

3. **Which real data, chosen under a fixed budget, most reduce the simulation-to-reality gap when used to correct both the digital twin and the model?** Existing methods correct either the model or the simulation, or both with different, unselected real data. Enforcing invariance does not target the joint error term of the domain adaptation bound and can increase it when simulated and real data require different actions, whereas bringing the simulation closer to reality can reduce it directly. *Hypothesis:* At an equal budget, using the same selected real data to correct both the twin and the model reduces the gap more than correcting either one alone. Candidate views are scored by the reconstruction uncertainty at their pose, by how much the predicted action changes when the twin is perturbed within it and, if an unlabelled real image is available, by the representation distance between the rendered and the real view. The selection will be compared with random and failure-driven selection over a range of budgets counted in real images for navigation and in real episodes or frames for manipulation, including unlabelled images.
4. **Do the methods, applied with unchanged settings, improve generalization in unseen scenes and in the other task, and can properties of the shift measured beforehand predict when they do?** *Hypothesis:* The attribution of the gap from the first question and its localization from the second, measured in a new scene or task from a reference reconstruction and a few paired views without applying the method, are more strongly rank-correlated across held-out scenes with the gap reduction a method achieves than simple indicators are, such as the base model's gap or the image-level discrepancy. This will be tested on many held-out navigation scenes with tests for dependent correlations and, exploratively, on the manipulation scenes and between the tasks.

8. Wkład spodziewanych wyników dla rozwoju dyscypliny naukowej / Contribution of the expected results to development of scientific discipline

Krótki opis spodziewanego wkładu wyników rozprawy doktorskiej dla rozwoju dyscypliny naukowej, w której realizowana jest rozprawa doktorska (maks. 1 strona, czcionka 11 pt., interlinia 1). / Brief description of the expected contribution of the results of the doctoral dissertation to the development of the scientific discipline in which the doctoral dissertation is carried out (at most 1 page, font 11, line spacing 1).

The proposed PhD dissertation aims to contribute to the development of the scientific discipline by addressing the generalization of deep learning models trained in simulations built from real data, which remains one of the main obstacles to applying machine learning in physical AI. The working assumption of the dissertation is that the simulation-to-reality gap is caused mainly by a limited set of task-relevant discrepancies. These may be identified among the reconstruction errors of a digital twin, localized inside the model, and reduced with a small amount of well-chosen real data. The methods are intended to be general, i.e., not tied to a particular robot, scene or task, and will be examined on robot navigation and robotic manipulation.

First, the research is intended to provide a better understanding of which reconstruction errors of a digital twin harm generalization, together with a method that varies the twin accordingly rather than uniformly. Second, it aims to develop a way of locating the stage of a deep model at which task information available for simulated inputs is lost for real ones, using probes with control tasks and causal interventions, and of reducing the gap by aligning simulated and real representations at that stage. Third, it aims to propose a rule for selecting a small amount of real data that corrects both the digital twin and the model. Finally, it will examine when such improvements carry over to unseen environments and between tasks. This knowledge could complement the currently dominant randomization heuristics with more principled and interpretable approaches.



Since transfer from simulation to reality is an instance of learning under distribution shift, the findings may also be relevant to machine learning in general, in particular to domain adaptation, representation learning and active learning. Apart from purely scientific results, the methods could reduce the cost of deploying learning-based robots in new environments, e.g., in service, logistics and inspection robotics. The source code is planned to be released, which could facilitate further research in the field.

9. Planowane metody badawcze / Planned research methods

Krótki opis metod badawczych, które są będą użyte w czasie realizacji rozprawy doktorskiej dla (maks. 2 strony, czcionka 11 pt., interlinia 1). / Brief description of the research methods that will be used during the implementation of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

The research will follow the methods commonly adopted in machine learning and robot learning projects. Since no established theory guarantees generalization from simulation to reality, the dominant part of the work will be empirical. Domain adaptation theory relates the error of a model in reality to its error in simulation, the discrepancy between the two distributions and the error of the best model on both. It is used as motivation, and the properties of the proposed methods will be analysed theoretically whenever possible.

The methods will be examined on two physical AI tasks, robot navigation and robotic manipulation, which share the reconstruction tools. Digital twins will be built from real images with neural scene reconstruction, e.g., 3D Gaussian Splatting [10], and coupled with simulators for navigation (e.g., Habitat) and manipulation (e.g., ManiSkill3). Models that map camera images to actions will be trained or fine-tuned in these simulators, starting from pretrained models where possible.

The research does not rely on a robot of its own, so the evaluation will use public data. For navigation, it will use public datasets of real indoor scenes with independent captures of the same scenes (e.g., ScanNet++). The twin will be built from one capture, and the other will provide real frames and a dataset-based reference simulation. This reference isolates the gap caused by reconstruction fidelity and leaves out differences in actuation and sensors. For manipulation, twins of real tabletop scenes will be built from images of public robot datasets, following the visual matching of SIMPLER [14], and compared with published real-robot results of the same policies. New models will also be evaluated in simulations whose parameters are hidden from the learner.

For the first research question, the research will examine variants of the twin in which one type of reconstruction error is corrected in selected regions using a more accurate reference, e.g., the laser scans of ScanNet++, or injected in a controlled way, to attribute the gap to types of error, and will study randomization of appearance and geometry whose per-region strength follows the estimated reconstruction error and task relevance, compared with uniform randomization of equal total strength. For the second question, linear probes with control tasks, trained on simulated hidden states and tested on paired real ones, will show at which stage task information is lost on real inputs. Causal interventions on hidden states will test whether this information is used, and a paired alignment loss attached at that stage will be compared with the same loss at the input, at the final features, at all stages and at layers chosen by existing criteria. For the third question, rules for selecting a small amount of real data will be compared at equal budgets with random and failure-driven selection. The selected data will refine the reconstruction of the twin and enter the training of the model, and correcting both will be compared with correcting only one. For the fourth question, the methods will be applied to held-out scenes of both tasks to check whether the attribution and localization measured before applying a method predict its gain in each scene better than simple predictors, using rank correlations compared with tests for dependent correlations.

The evaluation will rely on standard metrics, such as success rate, and on the simulation-to-reality gap, i.e., the difference between the performance of the same model in the twin and in reality or its reference, measured per held-out scene in both tasks and complemented where possible by open-loop measures on real frames. Methods will be compared with relevant baselines under comparable



training effort and real data. Statistical methods will be used to provide reliable outcomes, including repeated runs over seeds and scenes and statistical tests for comparisons, and ablation studies will determine the contribution of individual components. Data for training, model selection and final evaluation will be kept separate. If access allows, a real robot in a robotics laboratory of the university will be used to check the direction of selected results and the validity of the dataset-based proxy.

Experiments will be implemented mainly in Python with PyTorch, with source code managed in Git, and run on GPU clusters of the Department, WCSS and PLGrid. Code and data will be released whenever licences allow, following open science standards. The findings will be published at leading conferences from the ministerial list, such as ICRA, IROS, RSS and IEEE RA-L, as well as NeurIPS, ICML, ICLR and CVPR, and presented at smaller venues such as MLinPL.

10. Streszczenie popularnonaukowe / Abstract for general public

Streszczenie popularnonaukowe należy przygotować w dwóch językach: polskim i angielskim. Wersje językowe muszą być tożsame. Opis powinien być napisany językiem popularnonaukowym i zawierać cel rozprawy, opis planowanych badań naukowych oraz powodów, dla których podjęta została ta tematyka badawcza, a także najważniejsze spodziewane efekty (maks. 1 strona, czcionka 11 pt., interlinia 1). / The description for general public must be in Polish and in English. The language versions must be identical. The description should be written for general public and should include the doctoral dissertation goal, description of research, reasons for attempting a particular research topic and substantial results expected (at most 1 page, font 11, line spacing 1).

Streszczenie popularnonaukowe

Sztuczna inteligencja coraz częściej trafia do systemów fizycznych, takich jak roboty poruszające się po budynkach lub przenoszące przedmioty. Sterujące nimi modele uczenia głębokiego potrzebują ogromnej liczby przykładów, a zbieranie ich w prawdziwym świecie jest powolne, kosztowne i czasem niebezpieczne. Dlatego modele te uczą się zwykle w symulacji, a coraz częściej w cyfrowych bliźniakach, czyli wirtualnych kopiach rzeczywistych miejsc odtworzonych na podstawie ich nagrań. Taka kopia nigdy nie jest jednak idealna. Na przykład szklane drzwi lub odbicie w lustrze mogą zostać odtworzone błędnie i zmylić robota. W efekcie model, który dobrze działa w symulacji, w rzeczywistości często zawodzi, zwłaszcza w miejscach, których wcześniej nie widział. To jedna z głównych przeszkód w szerszym stosowaniu fizycznej sztucznej inteligencji. Celem rozprawy jest poprawa zdolności generalizacji modeli uczenia głębokiego trenowanych w cyfrowych bliźniakach. Generalizacja oznacza tu umiejętność poprawnego działania w warunkach innych niż te, w których model trenowano, a zwłaszcza w rzeczywistości. Badania w pierwszej kolejności ustalą, które różnice między symulacją a rzeczywistością faktycznie szkodzą modelowi i na którym etapie przetwarzania w modelu zamieniają się w błędy. Na tej podstawie powstaną metody, które nauczą model pomijać nieistotne różnice i więcej ćwiczyć tam, gdzie cyfrowy bliźniak jest najmniej dokładny. Powstaną też metody wyboru nielicznych rzeczywistych przykładów, które najlepiej poprawiają zarówno model, jak i cyfrowego bliźniaka. Metody zostaną sprawdzone w dwóch równorzędnych zadaniach, nawigacji robotów oraz manipulacji robotycznej (chwytniu i przestawianiu przedmiotów na stole), w środowiskach, których nie używano podczas uczenia. Wyniki uzyskane w symulacji zostaną porównane z opublikowanymi wynikami prawdziwych robotów, a jeśli to będzie możliwe, sprawdzone także na prawdziwym robocie. Zbadane zostanie też, czy poprawa uzyskana w jednym zadaniu przenosi się na drugie. Oczekiwany wynik rozprawy jest lepsze zrozumienie, dlaczego modele uczone w symulacji zawodzą w rzeczywistości, oraz metody, dzięki którym roboty będą mogły szybciej, taniej i bezpieczniej uczyć się pracy w nowych miejscach i przy więcej niż jednym zadaniu.

Abstract for general public

Artificial intelligence is increasingly used in physical systems, such as robots that move around buildings or carry objects. The deep learning models that control them need a huge number of examples, and collecting these in the real world is slow, costly and sometimes unsafe. Such models are therefore usually trained in simulation, and increasingly in digital twins, that is, virtual copies of



real places rebuilt from recordings of those places. Such a copy is never perfect, however. For example, a glass door or a reflection in a mirror may be rebuilt incorrectly and mislead the robot. As a result, a model that works well in simulation often fails in reality, especially in places it has not seen before. This is one of the main obstacles to wider use of physical artificial intelligence. The goal of the dissertation is to improve the generalization of deep learning models trained in digital twins. Generalization here means the ability to work correctly in conditions different from those the model was trained in, and in particular in reality. The research will first establish which differences between simulation and reality actually harm the model, and at which stage of processing inside the model they turn into errors. On this basis, methods will be developed that teach the model to ignore irrelevant differences and to practise more where the digital twin is least accurate. Methods will also be developed for choosing the few real examples that best correct both the model and the digital twin. The methods will be tested on two tasks of equal weight, robot navigation and robotic manipulation (grasping and moving objects on a table), in environments that were not used during training. The results obtained in simulation will be compared with published results of real robots and, where possible, also checked on a real robot. The research will also check whether an improvement obtained in one task carries over to the other. The expected outcome of the dissertation is a better understanding of why models trained in simulation fail in reality, and methods that let robots learn to work in new places and on more than one task faster, more cheaply and more safely.

11. Termin oddania do druku artykułu naukowego lub monografii / **Date of submission for publication of at least one scientific article or monography**

*Termin wysłania do recenzji artykułu naukowego lub monografii naukowej zgodnie z wymaganiami art. 186 ust. 1 pkt 3 Ustawy stawianymi osobie ubiegającej się stopień doktora. Termin dotyczy oddania do druku co najmniej jednego artykułu naukowego opublikowanego w czasopiśmie naukowym lub w recenzowanych materiałach z konferencji międzynarodowej w odpowiedniej dyscyplinie naukowej, które w roku opublikowania artykułu w ostatecznej formie były ujęte w wykazie („lista ministerialna”) sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. b) lub 1 monografii naukowej wydanej przez wydawnictwo, które w roku opublikowania monografii w ostatecznej formie było ujęte w wykazie sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. a). / **A date of submission for publication of at least one scientific article published in a scientific journal or in the peer-reviewed materials of an international conference in the relevant scientific discipline, which in the year of publication of the article in its final form were included in the list drawn up in accordance with the regulations issued under the Act (Art. 267 sec. 2 point 2(b) or 1 scientific monograph published by a publishing house, which in the year of publication of the monograph in its final form was included in the list drawn up in accordance with the regulations issued pursuant to the Act (Art. 267 sec. 2 point 2(a), referred to in Art. 186 sec. 1 point 3 of the Act.***

September 2027 (planned submission to a robotics conference or journal from the ministerial list, e.g. ICRA 2028 or IEEE Robotics and Automation Letters).

12. Inne / **Other comments**

*Dodatkowe wyjaśnienia, komentarze, podsumowanie związane z indywidualnym planem badawczym (maks. 1 strona, czcionka 11 pt., interlinia 1). / **Additional explanations, comments, summary related to the individual research plan (at most 1 page, font 11, line spacing 1).***

13. Określenie planowanej formy współpracy z promotorem / **Planned form of cooperation with the supervisor**



Krótki opis planowanej formy współpracy z promotorem, np. częstotliwość spotkań, forma spotkań i ich zakres, sposób komunikacji. / Brief description of the planned form of cooperation with the supervisor, e.g., frequency of meetings, form of meetings and their scope, method of communication.

It is planned to have weekly meetings with the supervisor. The meetings are planned to consist of results presentations, discussions, reviews and joint work on papers. The scope of meetings is also supposed to include the evaluation of progress and the planning of further activities. Moreover, it is planned to discuss results, issues and recent advancements in the field with other Ph.D. students weekly, with the supervisor's participation. Day-to-day communication will take place by e-mail and instant messaging.

14. Opinia promotora pomocniczego / Opinion of assistant supervisor

Promotor pomocniczy (jeżeli jest wyznaczony) powinien zaopiniować IPB. / Assistant supervisor (if appointed) should provide an opinion of the IRP.

nie dotyczy / not applicable

.....
Podpis promotora pomocniczego /
Signature of assistant supervisor

15. Podpisy doktoranta i promotora/promotorów / Signatures of doctoral student and supervisor(s)

Wybierz datę/Select date

.....
Data / Date

.....
Podpis doktoranta / Signature of doctoral student

.....
Podpis promotora / Signature of supervisor

.....
Podpis drugiego promotora / Signature of second supervisor